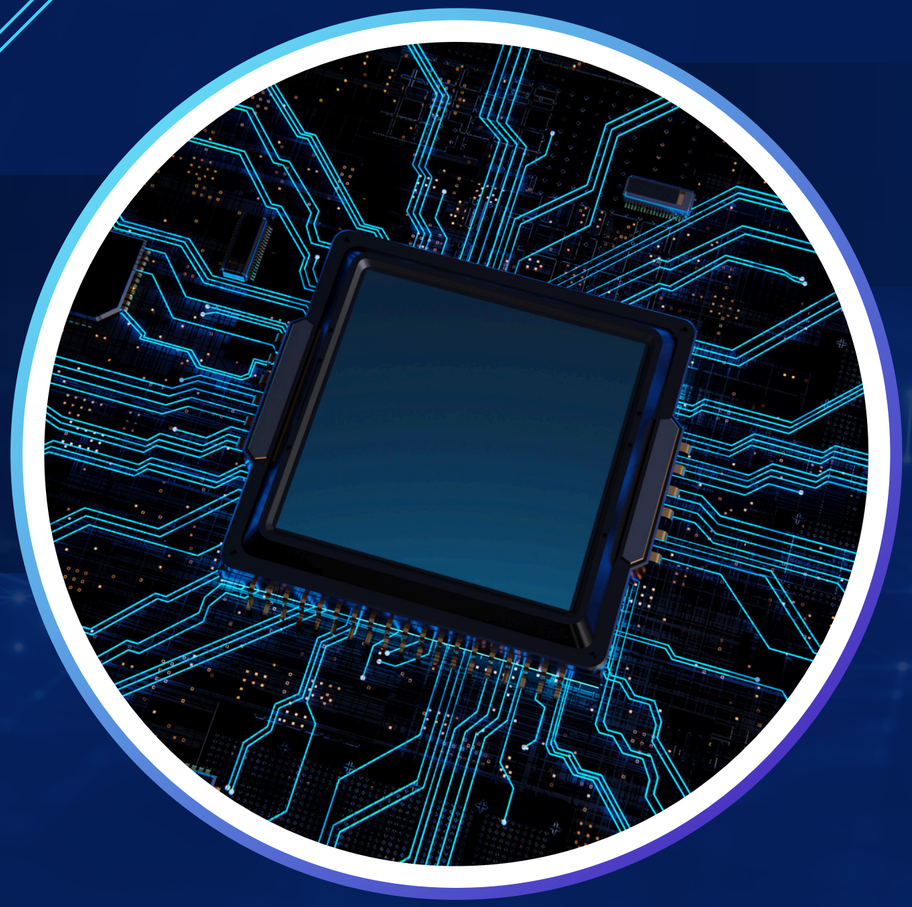
TYPE OF INTRUSION DETECTION SYSTEM



CLASSIFICATION

- Signature Vs Anomaly Based
- Active Vs Passive
- Host Vs Network
- Centralize vs Distributed
- Real Time Vs Interval

WAYS TO DETECT

Signature Recognition

- Catch the intrusions in terms of the characteristics of known attacks or system vulnerabilities.
- Best for: Detecting known attacks.
- Example: If a packet matches a known malware signature, the IDS triggers an alert.

Anomaly Detection

- Detect any action that significantly deviates from the normal behavior.
- Best for: Detecting unknown or new attacks.
- Example: A user logging in at 3 AM with unusually high data transfers may trigger an alert.

SIGNATURE RECOGNITION

What is it?

- Signature-based detection compares incoming data or actions against predefined patterns of known attacks

How does it work?

- Just like antivirus software matches virus definitions, a signature-based IDS.
- Detection occurs when a matching malicious behavior is observed

Analogy

- Imagine a security guard checking faces against a wanted poster. If someone matches the poster, the guard sounds the alarm. That's how signature detection works, by recognizing known attackers

What is Signature?

- Signature = Pattern of Code
- Patterns of known threats, like viruses, exploits, or malicious commands. If a match is found, the system raises an alert.

METHODOLOGY

ADVANTAGE

Known Threats

Detects threats precisely if a matching signature exists.

Low False Positive

Since it only alerts on known patterns, it rarely flags normal behavior as malicious.

Fast Detection

Immediate response once a known signature is matched.

DISADVANTAGE

Cannot Detect Unknown Attacks

Misses new or modified (zero-day) threats not in the database.

Requires Regular Updates

Needs continuous signature updates to stay effective.

Limited Flexibility

Only useful against previously identified threats.

ANOMALY RECOGNITION

What is it?

A method that identifies deviations from normal behavior in a system, network, or user activity. Once the baseline is established, it monitors system activity and raises alerts when it detects activity that significantly differs from this normal pattern.

How It Works?

- Learn Normal Behavior
- Monitor Current Activity
- Detect Deviations (A change or difference from what is normal, expected, or usual behavior.)

Analogy

Anomaly-based detection is like having a security guard who knows your daily routine. If something unusual happens, the guard may suspect something’s wrong even if he doesn't know exactly what.

METHODOLOGY

ADVANTAGE

Detects Unknown/Zero-Day Attacks

Identifies suspicious behavior even if the threat is new.

Monitors Behavioral Changes

Good for spotting insider threats or unusual user actions.

Adaptive to New Environments

Can adjust to new patterns over time with proper training.

DISADVANTAGE

High False Positive Rate

Normal but unusual activities can trigger alerts.

Requires Training Period

Needs time to learn what “normal” behavior looks like.

Complex Configuration

More challenging to maintain and fine-tune than signature-based systems.

COMPARISON

SIGNATURE

- Uses known attack signatures or patterns
- Very effective for known attacks
- Fast, since it matches against known patterns
- Easier to maintain; just update signatures

Snort, Suricata (using signature rules)

ANOMALY

- Detects deviations from normal behavior
- Good for detecting unknown or zero-day attacks
- Slower, due to analysis and behavior comparison
- Harder; needs tuning and baseline adjustment

ML-based IDS, systems that monitor behavior trends

TYPE OF INTRUSION DETECTION SYSTEM

Host Based IDS (HIDS)

- Installed on a computer to detect suspicious activity inside the system.

Network Based IDS (NIDS)

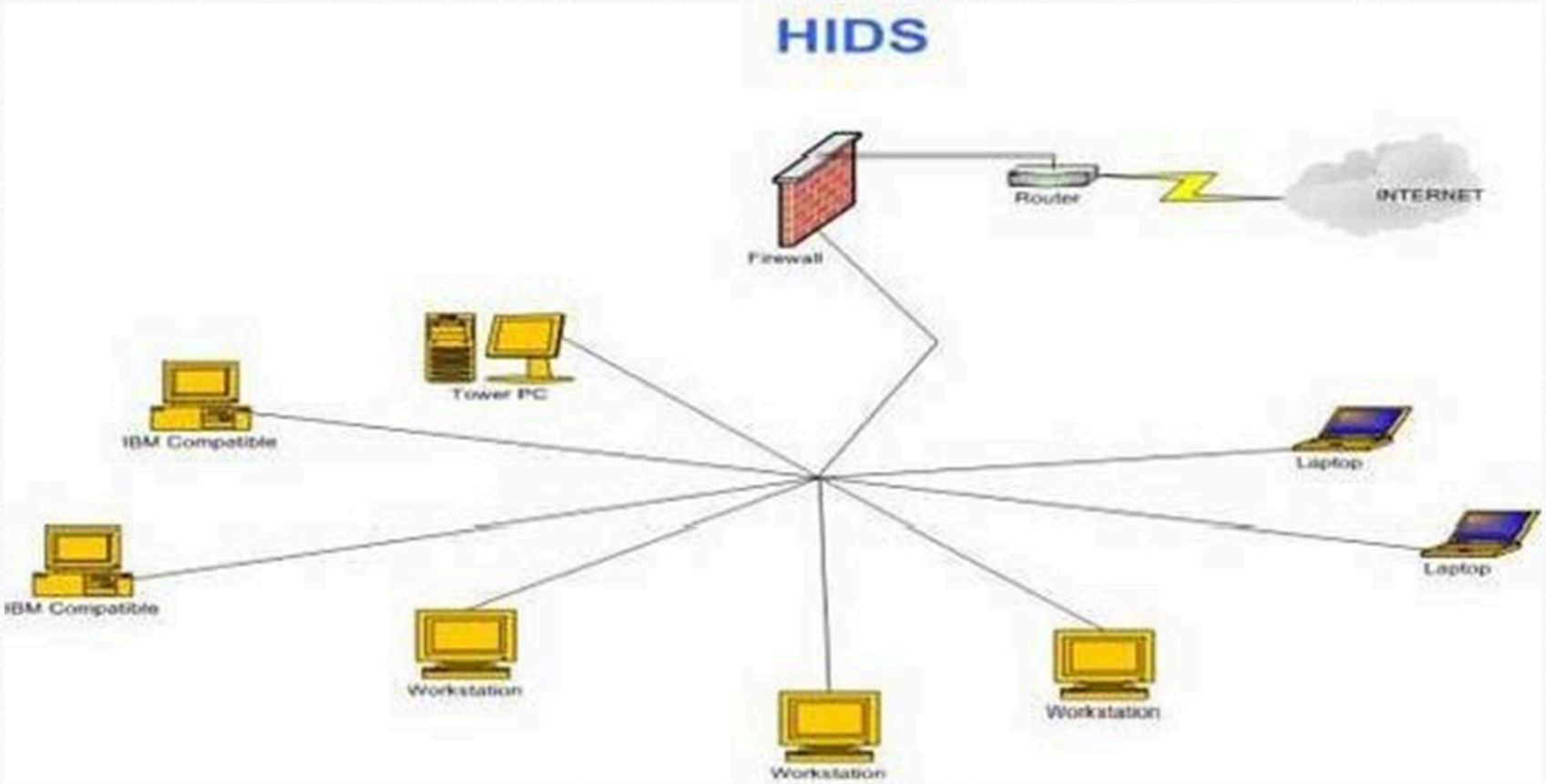
- Monitors network traffic to catch external threats.

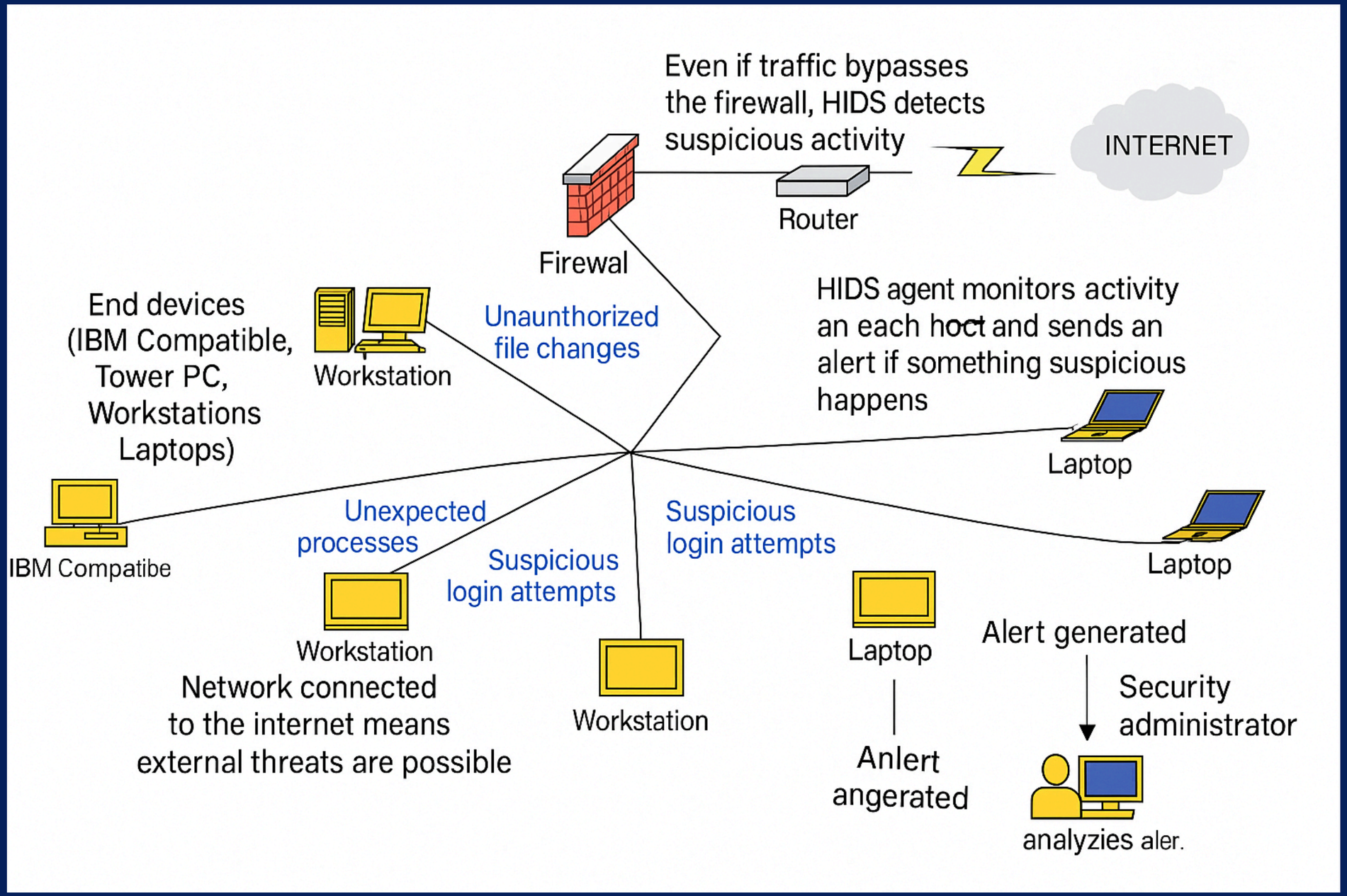
Hybrid

- Combines both HIDS and NIDS for stronger and more complete protection.

HOST-BASED IDS (HIDS)

- Host-based intrusion detection systems or HIDS are installed as agents on a host
- Host-based IDSs check for intrusions by checking information at the host or operating system level.
- These IDSs examine many aspects of your hosts, such as system calls, audit logs, error and messages logs. To detect any intruder activity.


HIDS
Firewall
Router
INTERNET
Tower PC
IBM Compatible
IBM Compatible
Laptop
Laptop
Workstation
Workstation
Workstation


Even if traffic bypasses the firewall, HIDS detects suspicious activity
INTERNET
Router
Firewal
End devices (IBM Compatible, Tower PC, Workstations Laptops)
Workstation
Unaunthorized file changes
HIDS agent monitors activity an each hoct and sends an alert if something suspicious happens
Laptop
Laptop
IBM Compatibe
Unexpected processes
Suspicious login attempts
Suspicious login attempts
Workstation
Network connected to the internet means external threats are possible
Workstation
Laptop
Alert generated
Security administrator
Anlert angerated
analyzies aler.

EXPLANATION OF THE DIAGRAM

Diagram Components

1) End Devices (Tower PC, Workstations, Laptops)

- Each of these computers has a HIDS agent installed.
- The agent monitors activity on that specific machine and sends alerts if something suspicious happens.

2) Firewall & Router

- These are part of the network perimeter but are not part of HIDS itself.
- Even though they filter traffic, HIDS works inside the hosts, not at the perimeter.

3) Internet Connection

- The diagram shows the network connected to the internet, meaning external threats are possible.
- HIDS will detect suspicious activity that has already bypassed the firewall or originated internally.

HOW THIS SETUP WORKS?

Step by step

HIDS agent installed on every device in the network.

HIDS Agent keep monitors activity

When the agent detects something suspicious:

- Generates an alert.
- Sends it to the central management system (if used) or directly to the security team.

ADVANTAGE

DETAILED HOST-LEVEL MONITORING

- Can detect changes in files, configurations, and logins inside the system.
- It can detect suspicious activities that network monitoring might miss especially those that happen entirely inside the host without involving network traffic.

CAN DETECT ENCRYPTED ATTACKS

- Many modern threats are hidden inside encrypted traffic (HTTPS, VPN, SSH).
- Works after traffic is decrypted, so it catches threats NIDS might miss.

ACCURATE USER ACTIVITY TRACKING

- HIDS can track which user accounts perform specific actions on the system.
- For example, if a privileged account creates or deletes system files at unusual times, HIDS can log and alert on that behavior.

DISADVANTAGE

NO NETWORK-WIDE VIEW

- HIDS can only protect and monitor the host it's installed on.
- If an attack is targeting other systems in the network, the HIDS won't see it.

CONSUMES HOST RESOURCES

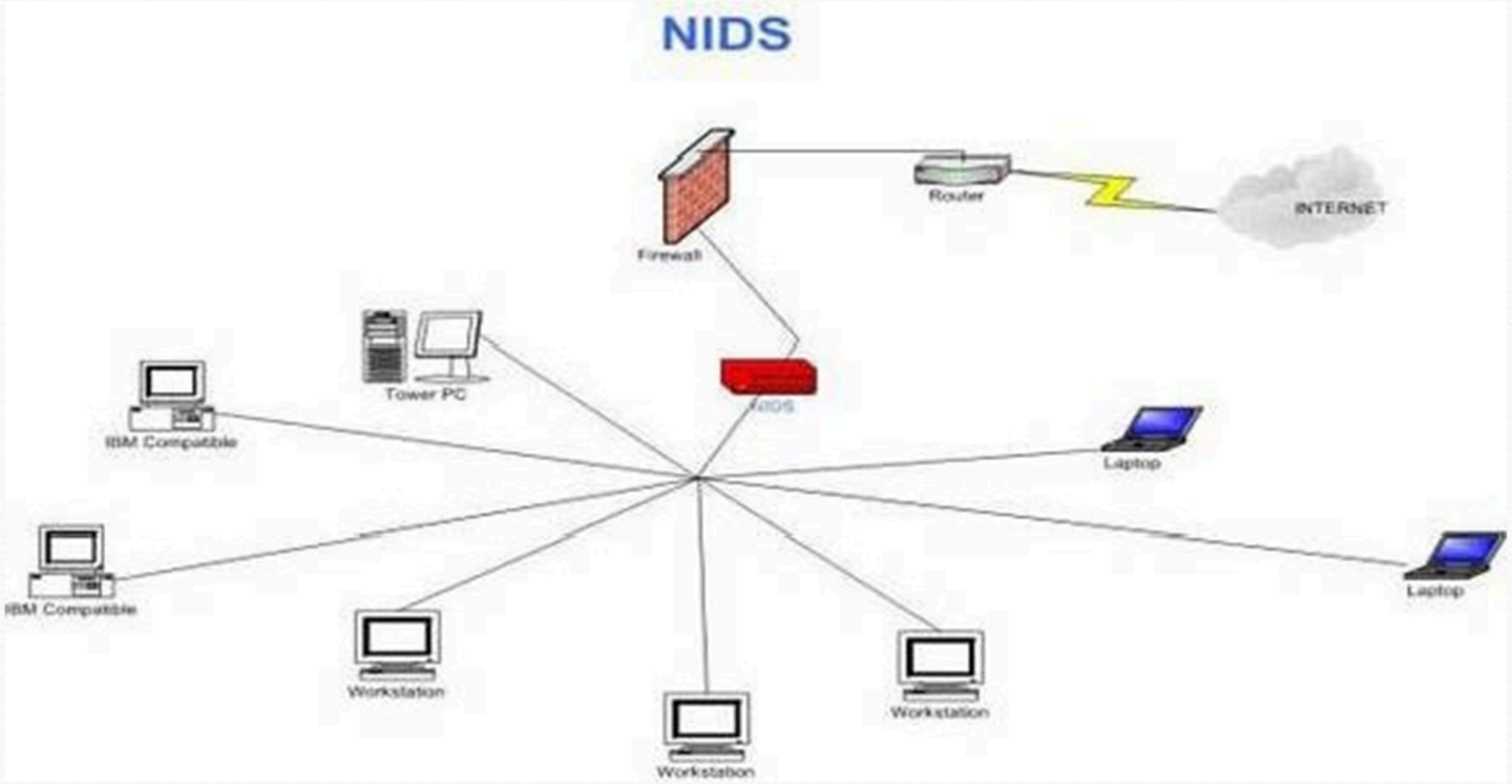
- Because HIDS is installed directly on the machine, it uses CPU, RAM, and storage to process logs, monitor activity, and maintain alerts.

VULNERABLE IF HOST IS COMPROMISED

- If an attacker gains full control of a host (root/admin access), they can disable, alter, or erase HIDS logs.
- This makes HIDS less effective if the system's integrity is already broken.

NETWORK BASED IDS (NIDS)

- NIDS (Network-based Intrusion Detection System) is a security tool that monitors and analyzes network traffic in real time to detect suspicious or malicious activity.
- It capture data packets traveling on the network media (cables, wireless) and match them to a database of signatures
- Depending upon whether a packet is matched with an intruder signature, an alert is generated or the packet is logged to a file or database.


NIDS
Firewall
Router
INTERNET
Tower PC
NIDS
IBM Compatible
Laptop
IBM Compatible
Laptop
Workstation
Workstation
Workstation

EXPLANATION OF THE DIAGRAM

Diagram Components

1) Internet → Router → Firewall
- Traffic from the internet first reaches the router, then goes through a firewall.
- The firewall filters unwanted traffic based on predefined rules.

2)NIDS Placement
- The NIDS device is positioned inside the network, typically behind the firewall.
- This allows it to monitor both inbound and outbound traffic for suspicious activity.
- It analyzes network packets to detect intrusions, malware, or policy violations.

3)Internal Network
- Multiple devices are connected to the network
- The NIDS monitors the traffic flowing to and from these devices without interfering (passive monitoring).

HOW THIS SETUP WORKS?

Step by step

Traffic Enters the Network

Data from the internet flows through:

- Router – directs the traffic to the correct destination.
- Firewall – blocks or allows traffic based on security rules.

NIDS Listen to Traffic

- The NIDS device is connected in a way that it can “see” all the data traveling across the network (inbound & outbound).
- It does not block traffic instead, it copies the packets for inspection.

Packet Analysis

NIDS analyzes both parts using:

- Signature-based detection
- Anomaly-based detection
- This analysis happens in real-time so threats can be flagged immediately

Detecting Suspicious Activity

Detecting Suspicious Activity Examples:

- A workstation sending data to an unknown foreign server at 3 AM.
- A laptop trying to run port scans on other devices.
- Malicious code hidden inside network packets.

ADVANTAGE

CENTRALIZED MONITORING

- Monitors the whole network from a single point without needing software on each device.
- It’s easier to maintain compared to installing software on every device.

NO PERFORMANCE IMPACT ON DEVICES

- NIDS operates on a separate device or server, not on the user’s machine.
- Users can work normally without even noticing the NIDS is there.

REAL-TIME DETECTION

- NIDS analyzes packets as they move across the network.
- Alerts are generated immediately when suspicious patterns are found.

DISADVANTAGE

CAN’T SEE ENCRYPTED TRAFFIC

- If traffic is encrypted (HTTPS, VPN, SSH), NIDS cannot inspect the packet contents.
- Malicious payloads inside encrypted streams may pass unnoticed.

MISSES LOCAL ATTACKS

- NIDS focuses on network traffic, so it can’t see what happens inside a single host.
- Attacks like local privilege escalation or file tampering may bypass detection.

FALSE POSITIVES

- Normal but uncommon activities may be flagged as suspicious.
- This can overwhelm administrators with unnecessary alerts.
- Over time, too many false positives may cause real threats to be ignored.

COMPARISON

HIDS

- Installed on a specific host (computer or server).
- Monitors activities of one host only.
- Uses system logs, file changes, and OS-level activity.
- Detects successful attacks on the host, even after they bypass the network.

Example: OSSEC, Tripwire, AIDE (Advanced Intrusion Detection Environment)

NIDS

- Placed at a strategic point in the network.
- Monitors traffic for the entire network segment.
- Uses packet captures and network traffic analysis.
- Detects attacks in transit before they reach hosts.

Example: Snort, Suricata, Bro/Zeek

HYBRID IDS

- A Hybrid Intrusion Detection System combines the features of Host-based IDS (HIDS) and Network-based IDS (NIDS) into one system.
- Correlates information from multiple sources, reducing false positives and false negatives compared to using HIDS or NIDS alone.
- Detects both external attacks before they reach hosts and internal threats that originate inside the network.

HOW IT WORKS?

Step by step

1) Data Collection
- The Hybrid IDS has network sensors (like NIDS) that capture packet data from the network.
- It also has host agents (like HIDS) installed on individual systems to monitor logs, file changes, and processes.

2) Centralized Analysis
- Both types of data are sent to a central management console.
- The system correlates the information. For example, if the NIDS detects a suspicious connection and the HIDS sees unusual file changes on the same host, it's likely a real attack.

3) Response
- Hybrid IDS can be passive (alert only) or active (block suspicious activity, similar to IPS).
- The system can automatically update detection rules to respond to new threats faster.

CLASSIFICATION

ACTIVE IDS

- Not just monitoring . It takes action when it detects threats
- Can block or modify network traffic
- Often called Intrusion Prevention Systems (IPS).

Snort (IPS mode), Suricata (IPS mode)

PASSIVE IDS

- Detects and alerts but does not take direct action.
- Requires admin or user to respond to threats.
- Works like a CCTV . Watches but can’t stop intrusions.

Zeek, OSSEC, Tripwire

CENTRALIZED IDS

- All monitoring and analysis happen in one central system.
- Easier to manage and maintain.
- May struggle with high-speed data streams.

Example: Cisco Secure IDS centralized console.

DISTRIBUTED IDS

- Uses multiple agents/sensors spread across locations.
- Better for large or complex networks.
- Requires secure communication between detection points.

Example: Wazuh with agents on multiple hosts.

REAL TIME IDS

- Detects and responds immediately.
- Suitable for stopping fast-moving threats.
- Critical for sensitive or high-value systems.
- Requires more processing power and continuous monitoring.

Snort, Suricata in Live Mode

INTERVAL IDS

- Analyzes data at scheduled times (hourly, daily).
- Slower — threats may already have occurred before detection.
- Good for environments where immediate action isn't critical.
- Threats may remain active before detection occurs.

AIDE or Tripwire on daily scan schedule

CONCLUSION

Intrusion Detection Systems (IDS) are an important part of network and system security. They can be classified in several ways:

- By detection method – Signature-based (detects known attacks quickly but cannot find new ones) and Anomaly-based (detects unusual activity and can find new threats, but may give false alarms).
- By location – Host-based IDS (HIDS) monitors activity on a single computer or server, while Network-based IDS (NIDS) monitors traffic across the network.
- By combination – Hybrid IDS uses features from different types to improve detection.
- By response type – Active IDS can take action (like blocking an IP), while Passive IDS only alerts the user.

For strong protection, organizations often use a mix of IDS types. This ensures they can detect both known and unknown attacks, protect both individual devices and entire networks, and respond effectively to threats. Understanding these types helps students design better security strategies in real-world situations.

THANK YOU FOR YOUR ATTENTION